Third dose of COVID-19 mRNA vaccine closes the gap in immune response between naïve nursing home residents and healthy adults

Pieter Pannus[1] • Stéphanie Depickère[1] • Delphine Kemlin[2] • Daphnée Georges[2,3] • Sarah Houben[1] • Véronique Olislagers J[2] • Alexandra Waegemans[2] • Stéphane De Craye[1] • Antoine Francotte[1] • Félicie Chaumont[1] • Celien Van Oostveldt[1] • Leo Hendrickx[4] • Johan Michiels[4] • Elisabeth Willems[4] • Emilie Dhondt [4] • Marharyta Krauchuk[4] • Marie-Noëlle Schmickler[5] • Mathieu Verbrugghe[5] • Nele Van Loon[5] • Katelijne Dierick[1] • André Matagne[3] • Isabelle Desombere[1] • Kevin K Ariën[4,6] • Arnaud Marchant[2] • Maria E. Goossens[1]

1. SD Infectious Diseases in Humans, Sciensano, Brussels, Belgium • 2 Institute for Medical Immunology and ULB Centre for Research in Immunology (U-CRI), Université´ Libre de Bruxelles (ULB), Gosselies, Belgium • 3. Laboratory of Enzymology and Protein Folding, Centre for Protein Engineering, InBioS, University of Liège, Liège, Belgium • 4.Virology Unit, Department of Biomedical Sciences, Institute of Tropical Medicine, Antwerp, Belgium • 5. Mensura EDPB, Occupational Health Service, Antwerp, Belgium • 6. Department of Biomedical Sciences, University of Antwerp, Antwerp, Belgium

CONCLUSION: A third dose of mRNA COVID-19 vaccine closes the gap in humoral and cellular immune response observed after primary vaccination between nursing home residents and staff members but suggest that further boosting might be needed to achieve optimal protection against variants of concern in this vulnerable population.

Nursing home residents, a frail and old population group, respond poorly to primary mRNA COVID-19 vaccination. A third dose has been shown to boost protection against severe disease and death in this immunosenescent population, but limited data is available on the immune responses it induces.

Methods

- 85 nursing home residents and 88 staff members, dually vaccinated and naïve to SARS-CoV-2 infection, received a 3rd dose of mRNA COVID-19 vaccine.
- SARS-CoV-2 specific humoral and cellular immune responses were evaluated at day 0, day 28, and month 6 post third dose administration.
- Breakthrough infections were monitored in the six months following third dose administration.
- Analyses were performed on two cohorts: the complete cohort (n=173), and the immunogenicity cohort (n=84) for which complete data were available for day 28 post dose 3 as well as post dose 2.

Table 1. Demographics, characteristics by study groups

		Complete cohort (n = 173)		Immunogenicity cohort (n = 84)	
Characteristic		Staff (n=88)	Residents (n=85)	Staff (n=42)	Residents (n=42)
Age (years, mean ± SD)		48 ± 11	83 ± 11	47 ± 10	82 ± 12
Sex	Female	74 (84%)	46 (54%)	33 (79%)	23 (55%)
Ethnicity	White	85 (97%)	83 (98%)	40 (95%)	40 (95%)
BMI (mean ± SD)		27.1 ± 5.9	25.1 ± 5.1	26.3 ± 5.7	24.6 ± 5.1
Self-reported smoking status	Former smoker	5 (5%)	6 (7%)	3 (7%)	2 (5%)
	Nonsmoker	70 (80%)	69 (81%)	32 (76%)	33 (78%)
	Current smoker	13 (15%)	10 (12%)	7 (17%)	7 (17%)
Comorbidity (at least one self-reported)	Yes	5 (6%)	55 (65%)	3 (7%)	30 (71%)
Time lapse between dose 1 and 3 (days) (mean ± SD)		253 ± 18	247 ± 36	238 ± 11	249 ± 15

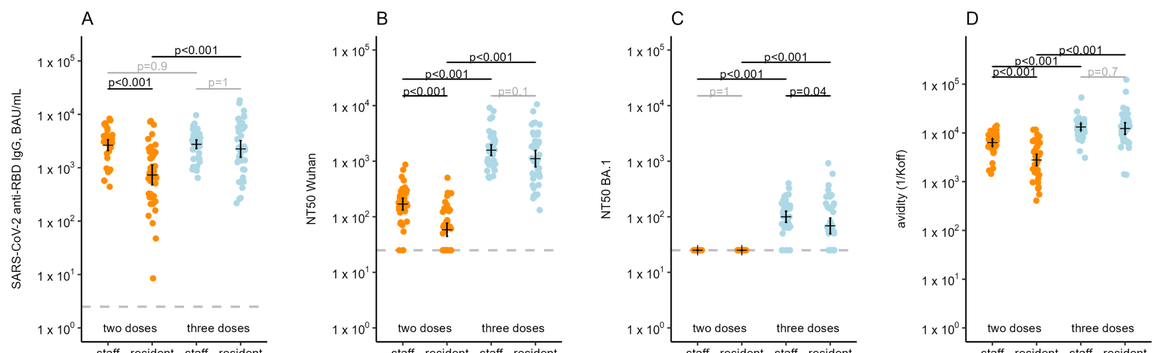

Fig. 1. Humoral immune responses after two (orange) and three (blue) vaccine doses in staff and residents (immunogenicity cohort). Black bars indicate GMT with 95% CI.

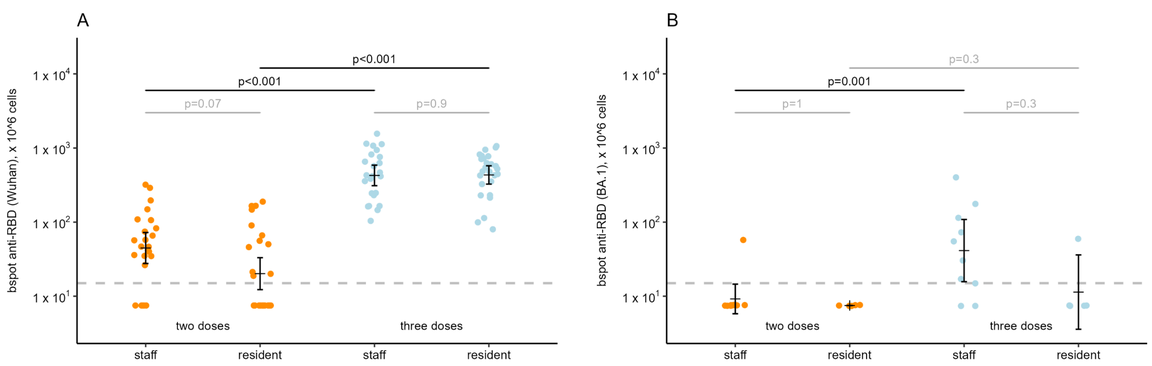

Fig. 2. Frequency of memory B cells (MBC) after two (orange) and three (blue) vaccine doses in staff members and residents of nursing homes, measured by B cell ELISpot and expressed as Spot Forming Cells (SFC) per million input cells, for Wuhan and Omicron BA.1. Black bars indicate GMT with 95% CI.

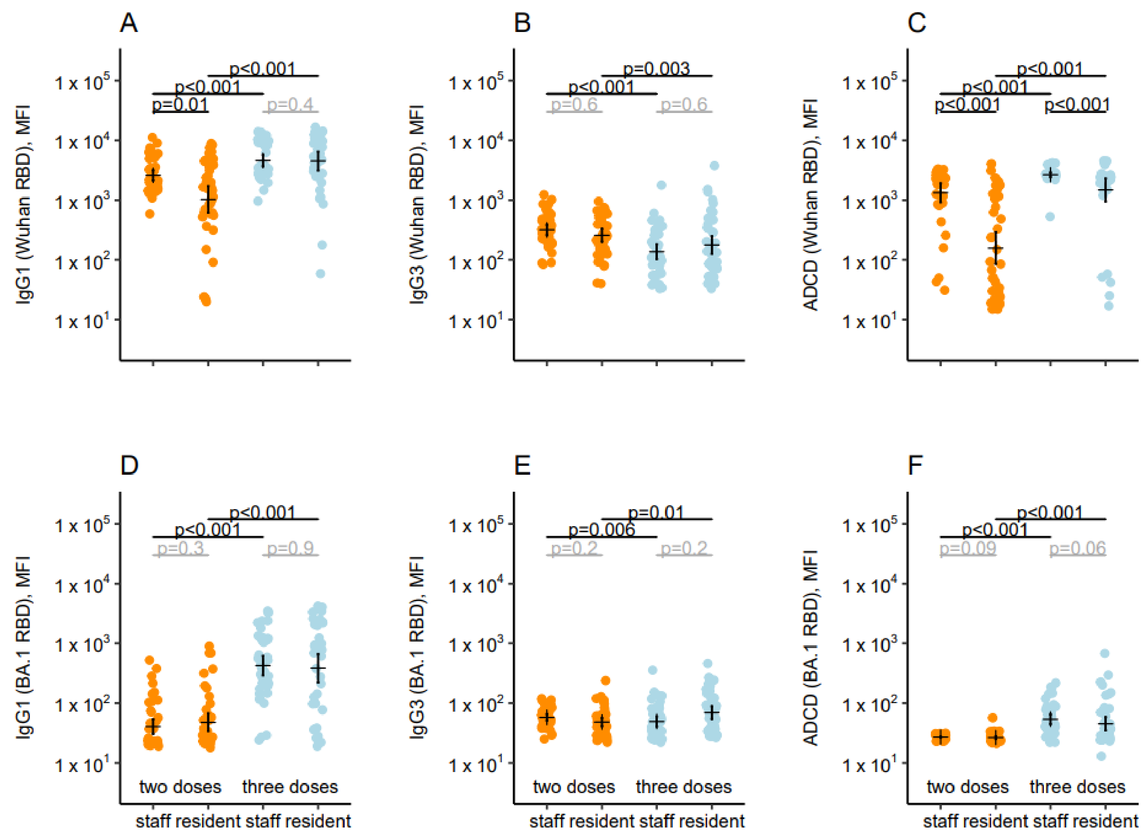

Fig. 3. IgG subclasses and antibody dependent complement deposition (ADCD) measured for RBD Wuahn and Omicron BA.1. Black bars indicate GMT with 95% CI..

Results

- 28 days post third dose, levels of **binding** and **neutralizing** antibodies, and antibody **avidity**, normalized between staff and residents (fig.1).
- All residents had detectable wild type specific neutralizing antibodies post dose 3, compared to just 57% post dose 2.
- **Omicron**-specific neutralizing antibodies remained slightly lower in residents compared to staff.
- A similar normalization was observed in levels of **IgG subtypes** 1 and 3, and in levels of **antibody dependent complement deposition**.
- **Memory B cells** specific for wild type virus were detectable in all participants after third dose. BA.1 specific responses were much lower and only detectable in staff.
- **Breakthrough infections,** in the following six months post third dose were detected in 47% of residents and 49% of staff but were all pauci-symptomatic.
- Incidence of breakthrough infection was **correlated** with levels of SARS-CoV-2 specific IgGs, neutralizing antibodies, antibody avidity and ADCD responses.

REFERENCES

- Pannus, P. Depickère S. et al. Third dose of COVID-19 mRNA vaccine closes the gap in immune response between naïve nursing home residents and healthy adults. Vaccine, 2023, 41 (17): 2829-2836.

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Sciensano • Stéphanie Depickère • T + 32 2 642 57 92 • stephanie.depickere@sciensano.be • **www.sciensano.be**